Chapter 5

Two-Stage CMOS Operational Amplifier

5.1 Introduction

In single stage Op-Amp, small signal current produced by the input pair to flow directly through the output impedance. The gain of these topologies is limited by the product of the input pair transconductance and the output impedance. In cascode circuit, its provide high gain while limiting the output swing. In some application, op-amp requires high gain and large output swing with operate 0.5V power supply. In such a case, we require extra stage so called two-stage. In multistage amplifier, first stage is require to provide a high resistance source. In differential amplifier, input stage must provide large CMRR. The function of the middle stage of an amplifier cascade is to provide the bulk of the voltage gain, also provide other function like conversion of the signal from differential mode to single-ended mode and shifting of the dc level of the signal in order to allow the output signal to swing both positive and negative. The main function of the last stage of an amplifier is to provide a low output resistance in order to avoid loss of gain when a low-valued load resistance is connected to the amplifier, also output stage should be able to supply the current required by the load that is without dissipating large amount of power in the output transistors.

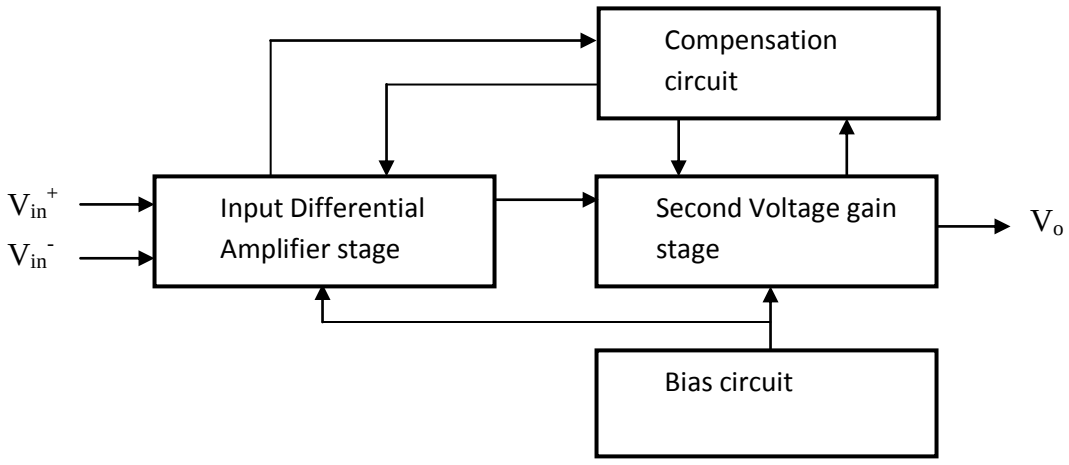


Figure 5.1: Block diagram of Two Stage CMOS Op-Amp [1]

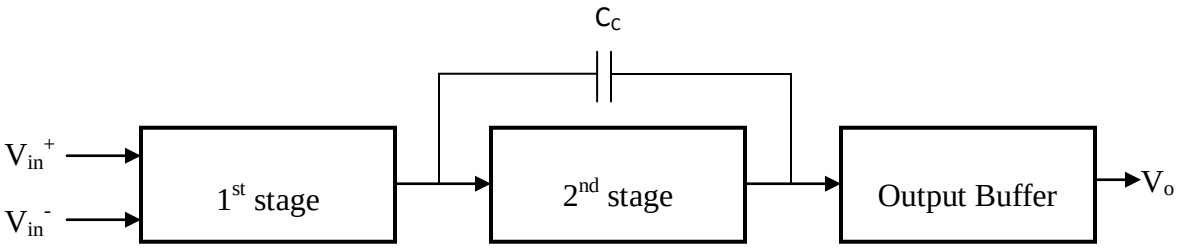


Figure 5.2: Simplified block diagram of Two-Stage Op-amp [1]

Basically operational amplifier has three stages, but ignoring output buffer stage so called "two-stage" op-amp. "Two-stage" refers as number of gain stages in the op-amp.

Operational Amplifiers are the backbone for many analog circuit designs. The speed and accuracy of these circuits depends on the bandwidth and DC gain of the Op-amp. Larger will be the bandwidth and gain, higher will the speed and accuracy of the amplifier. The general block diagram of an op-amp is shown above in Figure 5.1. The simplified Block Diagram is shown in figure 5.2.

The first block is a differential amplifier. It has two inputs that are the inverting and non-inverting voltage. It gives a differential voltage at the output or a differential current which depends only on differential input voltage. The second block is a differential ended to single-ended converter. It is used to transform the differential signal generated by the first block into a single ended output signal. Some architecture doesn't require the differential to single ended function; therefore this block can be eliminated in that. In single stage amplifier, the gain provided by the input stages is not sufficient, so there is an additional amplification is required which is provided by the second stage, i.e. the common source amplifier, driven by the first stage output. As this stage uses differential input unbalanced output differential amplifier, so it provides the required extra gain. The biasing circuit is here to provide the proper operating point to each transistor in its saturation region. Third stage, the output buffer stage provides the low impedance at output and larger output current needed to drive the load of op-amp or improves the slew rate. Even the output stage can be dropped since many applications do not need low output impedance. If the op-amp is intended to drive a small purely capacitive load, then output buffer is not required. Compensation capacitor is included to ensure stability when

op-amp is used with feedback. Because C_C is place between input and output of the second common source stage, so it is called Miller capacitance.

5.2 Two-Stage CMOS Op-Amp

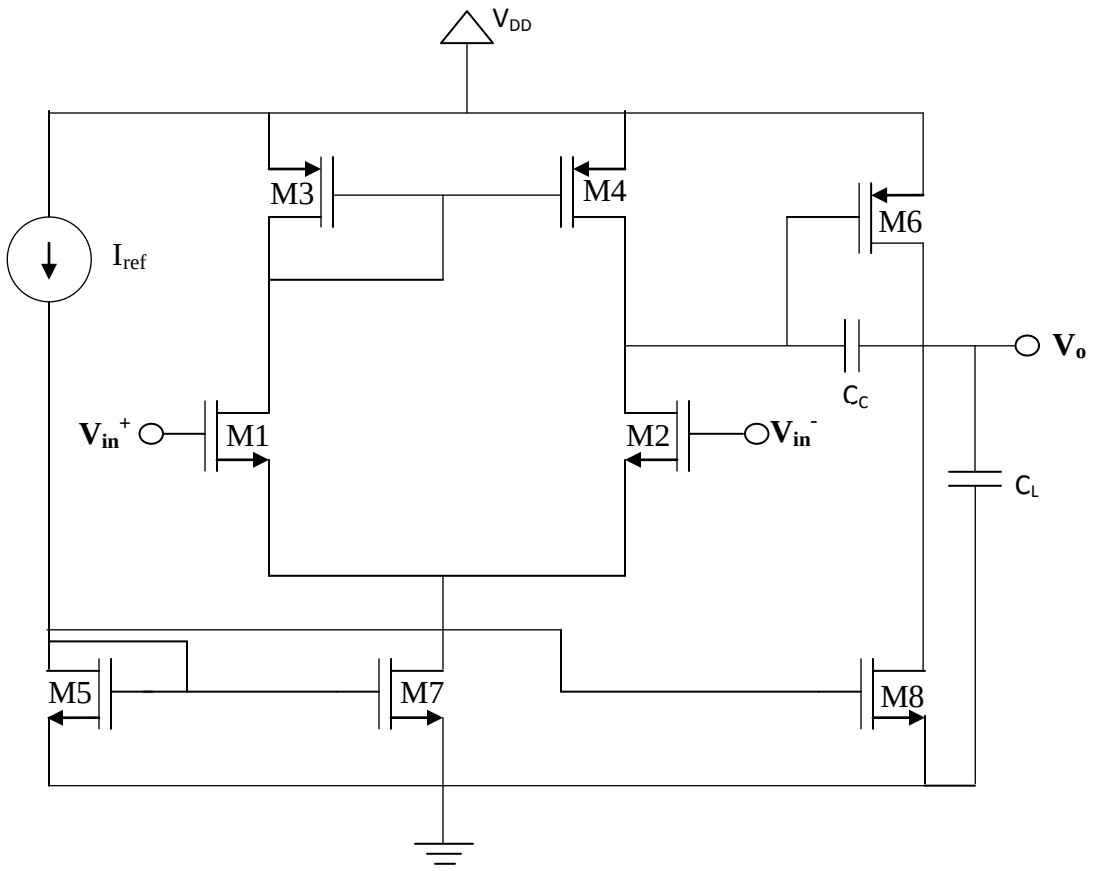


Figure 5.3: Two-Stage Op-Amp using miller compensation

The topology of this circuit is that of a standard two-stage CMOS op-amp. it has three subparts, namely input differential amplifier stage, second gain stage and biasing circuit.

5.3 Circuit Description

Two-stage op-amp is containing three subsections:

First Stage: Figure 5.3 contains two part of two-stage op-amp. Transistor M1, M2, M3, and M4 constitute the first stage of differential amplifier. M1 and M2 transistors are nMOS. M3 and M4 transistors are pMOS transistor. Input voltage V_{in}^{+} and V_{in}^{-} are given to gate terminal of transistor M1 and M2. Gate of M1 is noninverting input and gate of M2 is inverting input. M3 and M4 constitute current mirror. A differential input signal is

applied across the two input terminals will be amplified according to the gain of the differential stage. So that gain of this stage is the transconductance of M1 times to the total output resistance seen at the drain of M4. The main resistances that constitute to the output resistance are that of the input transistors and output resistance of the load transistors M3 and M4. Transconductance and output resistance of the first stage is given by G_{m1} and R_{out1}

$$G_{m1} = g_{m1} \tag{5.3.1}$$

$$R_{out1} = r_{o4} \| r_{o2} \tag{5.3.2}$$

In this circuit, current mirror active load used has three main advantages. First, active load devices crate large output resistance that provides small amount of chip area. Second current mirror topology provides the differential to single ended conversion of input signal. The voltage gain of the differential amplifier is given by A_{V1}

$$A_{V1} = -g_{m1} \times (r_{o4} \| r_{o2}) \tag{5.3.3}$$

The differential to single ended conversion is achieved by using current mirror M3 and M4. The current from M1 is mirror by M3 and M4 and subtracted from the current from M2. Finally the differential current from M1 and M2 multiplied by the output resistance of the input stage that gives the single ended output voltage. This is the part of the second stage.

Second Stage: Second stage is a common source stage. The aim of this second stage is to provide the additional gain consisting of transistors M6 and M8. Input of the second stage is the output of the differential amplifier stage. This stage receives the output from the drain of M4 and amplifies it through M6 by the common source configuration. Similar to differential amplifier, this stage employs an active device, M8 to work as the load resistance for M6. The gain of this stage is the transconductance of M6 times the effective load resistance comprised of the output resistance of M6 and M8. Transconductance and output resistance of the second stage is given by G_{m2} and R_{out}

$$G_{m2} = g_{m6} \tag{5.3.4}$$

$$R_{out} = (r_{o6} \| r_{o8}) \tag{5.3.5}$$

M6 is driver and M8 is acts as the load. The voltage gain of the second stage is given by A_{V2}

$$A_{V2} = -g_{m6} \times (r_{o5} \| r_{o8}) \tag{5.3.6}$$

So that total gain of the two stage CMOS operational amplifier is given by

$$A_V = A_{V1} \times A_{V2} \tag{5.3.7}$$

$$A_V = (-G_{m1} R_{out\,1}) \times (-G_{m2} R_{out}) \tag{5.3.8}$$

$$A_V = g_{m1}(r_{o4} \| r_{o2}) g_{m6}(r_{o6} \| r_{o8}) \tag{5.3.9}$$

Biasing Circuit: The purpose of the biasing circuit is to provide the gate to source voltage of the entire transistor. The biasing of the operational amplifier is achieved with only three transistors. Transistor M5 supplies a voltage between gate and source of M7 and M8. Transistors M7 and M8 sink a certain amount of current based on their gate to source voltage which is controlled by the bias string. M5 is diode connected to ensure they operate in the saturation region. Proper biasing of the other transistor in the circuit (M1-M4, M6) is controlled by the node voltages present in the circuit itself. M6 is biased by the gate to source voltage set up by the V_{GS} of the current mirror load as are the M3 and M4 transistors.

Third stage: In fig 5.2, third stage is output buffer stage which is simply common drain stage. This stage is often called a source follower, because the source voltage follows the gate voltage, except for a level shift. When possible, it is desirable to tie the substrate of the source-follower device to its source in order to eliminate gain degradations due to the body effect. This connection also results in a smaller dc voltage drop from the gate to the source of the source-follower device, which is a major limitation on the maximum positive output voltage. So that in fig 5.3, we don't show output buffer stage due above drawback.

5.4 Small single analysis

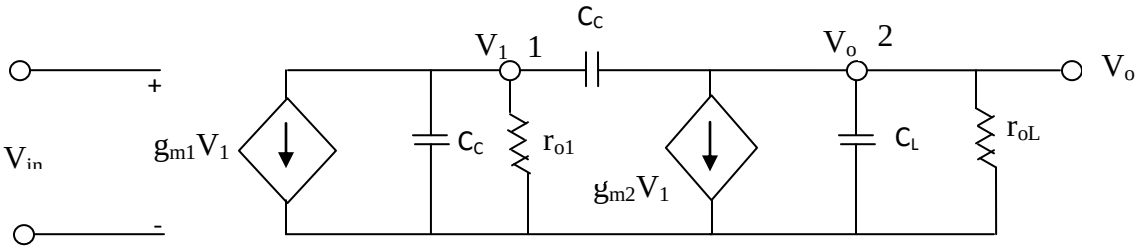


Figure 5.4: Small signal analysis of two-stage CMOS op-amp without miller compensation

General transfer function of fig. 5.4 is given by $\text{T.F} = \frac{V_o}{V_{in}}$

As per small signal diagram, $$\frac{V_o}{V_{in}} = \frac{V_1}{V_{in}} \times \frac{V_o}{V_1} \tag{5.4.1}$$

Apply KCL to node 1 above circuit,

$$\frac{v_1}{\frac{1}{sC_1}} + \frac{v_1}{R_1} + g_{m1}v_{in} + \frac{v_1 - v_o}{\frac{1}{sC_L}} = 0$$

$$v_1 sc_1 + \frac{v_1}{R_1} + g_{m1}v_{in} + sc_c(v_1 - v_o) = 0$$

$$v_1\left(sc_1 + \frac{1}{R_1} + sc_1\right) + g_{m1}v_{in} - v_o sc_c = 0$$

$$v_1 = \frac{v_o sc_1 R_1 - g_{m1}v_{in}R_1}{1 + sR_1(c_c + c_1)} \tag{5.4.2}$$

Apply KCL to node 2 in above circuit,

$$\frac{v_o}{\frac{1}{sc_2}} + \frac{v_o}{R_2} + g_{m2}v_1 + \frac{v_o - v_1}{\frac{1}{sc_c}} = 0$$

$$sc_2 v_o + \frac{v_o}{R_2} + g_{m2}v_1 + sc_c(v_o - v_1) = 0$$

$$v_o\left[sc_2 + \frac{1}{R_2} + sc_c\right] - v_1 sc_c + g_{m2}v_1 = 0$$

$$v_o\left[\frac{1}{R_2} + (sc_2 + sc_c)\right] = v_1[sc_c - g_{m2}]$$

$$v_o[1 + sR_2(c_c + c_2)][1 + sR_1(c_c + c_1)] = (v_o sc_c R_1 - g_{m1}R_1 v_{in})(sc_c - g_{m2})$$

$$\frac{v_o}{v_{in}} = \frac{g_{m1}g_{m2}R_1R_2\left(1-\frac{sc_c}{g_{m2}}\right)}{s^2[R_1R_2(c_1c_2+c_1c_c+c_2c_c)]+s[R_2(c_c+c_2)+R_1(c_c+c_1)+c_cg_{m2}R_1R_2]+1} \quad (5.4.3)$$

Equation 5.4.3 is standard transfer function of fig. 5.3.

Two pole transfer function:

$$\frac{v_o}{v_{in}} = \frac{A_{DC}\left(1-\frac{s}{Z}\right)}{\left(1+\frac{s}{P_1}\right)\left(1+\frac{s}{P_2}\right)} \quad (5.4.4)$$

$$\frac{v_o}{v_{in}} = \frac{A_{DC}\,(1-\frac{s}{Z})}{1+s\left(\frac{1}{P_1}+\frac{1}{P_2}\right)+s^2(\frac{1}{P_1P_3})} \quad (5.4.5)$$

$$s\left(\frac{1}{P_1}+\frac{1}{P_2}\right) = s\times\frac{1}{P_1} \quad (5.4.6)$$

$\frac{1}{P_2}$ =Co-efficient of 's'

If pole P_1 and P_2 is shown in graph. In this graphs pole P_2 is far away from the pole P_1 at high frequency so that pole P_1 is located at low frequency so that pole P_2 is ignored in equation (5.4.6).

Compare equation (5.4.3) and (5.4.5), poles P_1, P_2 , zeros Z and DC gain A_{DC} is given by

Pole P_1,

$$P_1 = \frac{1}{R_2(c_c+c_2)+R_1(c_c+c_1)+g_{m2}R_1R_2}$$

$$\boxed{P_1 \cong \frac{1}{g_{m2}R_1R_2c_c}} \quad (5.4.7)$$

Pole P_2,

$$P_1P_2 = \frac{1}{R_1R_2(c_1c_2+c_1c_c+c_2c_c)}$$

$$P_2 = \frac{1}{P_1R_1R_2(c_1c_2+c_1c_c+c_2c_c)}$$

$$P_2 = \frac{g_{m2}R_1R_2c_c}{R_1R_2(c_1c_2+c_1c_c+c_2c_c)} \quad (P_1 \text{ from equation 5.4.7})$$

$$P_2 = \frac{g_{m2}c_c}{(c_1c_2+c_1c_c+c_2c_c)}$$

$$P_2 \cong \frac{g_{m2}c_c}{c_cc_2}$$

$$\boxed{P_2 \cong \frac{g_{m2}}{c_2}} \quad (5.4.8)$$

Zeros Z,

$$\boxed{Z = \frac{g_{m2}}{c_c}} \quad (5.4.9)$$

DC voltage gain, $$\boxed{A_{DC} = g_{m1}g_{m2}R_1R_2} \quad (5.4.10)$$

5.5 Derivation of two-stage Op-Amp parameter

We will derive the performance parameter of two-stage CMOS operational amplifier which is given below.

- Gain
- Slew-rate, SR
- Gain-bandwidth product, GBW
- Phase margin, PM
- Output voltage swing
- Load capacitance, C_L
- Power dissipation, P_{diss}

Gain-bandwidth Product, GBW:

$$\text{GBW} = \text{DC gain} \times P_1$$

From the equation (5.4.10) and (5.4.7), put the value of DC gain and pole P_1

$$\text{GBW} = g_{m1}R_1 \cdot g_{m2}R_2 \times \frac{1}{g_{m2}R_1R_2C_C}$$

$$\text{GBW} = \frac{g_{m1}}{C_C} \quad (5.5.1)$$

Zeros is far away from the 10 times located after unity gain. So therefore, we are express zeros in bode plot diagram, $Z \geq 10\text{GBW}$

Slew-rate, SR: It is defined by the maximum rate of change of output voltage per unit time, generally express in V/μs

$$\text{SR} = \frac{dv}{dt}\Big|_{\text{Maximum}}$$

Relation between voltage and capacitor in term of capacitor

$$V = \frac{1}{C}\int I\, dt$$

$$VC = \int I\, dt$$

Replacing capacitor C by c_c and current I by I_o, $I_o = c_c\frac{dv}{dt} \Longrightarrow \frac{I_o}{c_c} = \frac{dv}{dt}$

Final equation of the slew rate is given by $\boxed{SR = \frac{I_0}{c_c}}$ (5.5.2)

Phase margin, PM:

From the equation (5.4.4), phase angle of the pole and zero of the overall transfer function is given by

$$\text{Angle } \frac{V_0}{V_{in}} = -\tan^{-1}\frac{W}{Z} - \tan^{-1}\frac{W}{P_1} - \tan^{-1}\frac{W}{P_2} \quad (5.5.3)$$

$$\frac{V_0}{V_{in}} = -\tan^{-1}\frac{GBW}{Z} - \tan^{-1}\frac{GBW}{P_1} - \tan^{-1}\frac{GBW}{P_2}$$

Putting the value of the GBW, Z, P_1 and P_2 from the equation (5.5.1) (5.4.7), (5.4.8) and (5.4.9)

$$\frac{V_0}{V_{in}} = -\tan^{-1}\frac{GBW}{10\,GBW} - \tan^{-1}\frac{g_{m1}g_{m2}R_1R_2C_C}{C_C} - \tan^{-1}\frac{GBW}{P_2}$$

$$\frac{V_0}{V_{in}} = -\tan^{-1}\frac{1}{10} - \tan^{-1} g_{m1}g_{m2}R_1R_2 - \tan^{-1}\frac{GBW}{P_2}$$

From the equation (5.4.10), $g_{m1}g_{m2}R_1R_2$ is voltage gain A_{DC}

$$\frac{V_0}{V_{in}} = -\tan^{-1}\frac{1}{10} - \tan^{-1} A_{DC} - \tan^{-1}\frac{GBW}{P_2}$$

Putting $\tan^{-1}\frac{1}{10} = 5.71$ and voltage gain is always 90° phase angle in bode plot

$$-180° + PM = -5.71 - 90° - \tan^{-1}\frac{GBW}{P_2}$$

$$PM = 180° - 5.71 - 90° - \tan^{-1}\frac{GBW}{P_2}$$

$$PM = 84.29° - \tan^{-1}\frac{GBW}{P_2} \quad (5.5.4)$$

Op-amp has better phase margin and improve stability at 60° phase angle. So that we will take phase margin is 60° as according to find relation between GBW and second pole P_2 in bode plot.

$$60° = 84.29° - \tan^{-1}\frac{GBW}{P_2}$$

$$\tan^{-1}\frac{GBW}{P_2} = 24.29°$$

$$\frac{\mathrm{GBW}}{\mathrm{P_2}} = \tan 24.29$$

$$\frac{\mathrm{GBW}}{\mathrm{P_2}} = 0.4513$$

$$\mathrm{P_2} = 2.2\ \mathrm{GBW}$$

For $\mathrm{PM} = 60°$, relation between P_2 and GBW is given by

$$\boxed{\mathrm{P_2} \geq 2.2\ \mathrm{GBW}} \qquad (5.5.5)$$

From the equation (5.5.4)

$$\mathrm{PM} = 84.29° - \tan^{-1}\frac{\mathrm{GBW}}{\mathrm{P_2}}$$

Now second case, take the phase margin PM = 45°

$$45° = 84.29° - \tan^{-1}\frac{\mathrm{GBW}}{\mathrm{P_2}}$$

$$\tan^{-1}\frac{\mathrm{GBW}}{\mathrm{P_2}} = 39.29°$$

$$\frac{\mathrm{GBW}}{\mathrm{P_2}} = 0.8181$$

$$\mathrm{P_2} = \frac{\mathrm{GBW}}{0.8181}$$

For, $\mathrm{PM} = 45°$, relation between P_2 and GBW is given by

$$\boxed{\mathrm{P_2} \geq 1.22\ \mathrm{GBW}} \qquad (5.5.6)$$

From the equation (5.4.8) and (5.5.1), put the value of P_2 and GBW in equation (5.5.5)

$$\frac{\mathrm{g_{m2}}}{\mathrm{C_2}} \geq 2.2\ \frac{\mathrm{g_{m1}}}{\mathrm{C_C}}$$

C_2 is output capacitance; we replace C_2 by load capacitance

$$\frac{\mathrm{g_{m2}}}{\mathrm{C_L}} \geq 2.2\ \frac{\mathrm{g_{m1}}}{\mathrm{C_C}} \qquad (5.5.7)$$

In transfer function, zeros is given by

$$\mathrm{Z} = 10\mathrm{GBW}$$

Form the equation (5.4.9) and (5.5.1)

$$\frac{g_{m2}}{C_C} = 10\ \frac{g_{m1}}{C_C}$$

$$g_{m2} = 10 g_{m1} \tag{5.5.8}$$

Put the value of g_{m2} in equation (5.5.7)

$$\frac{10\ g_{m1}}{C_L} \geq 2.2\ \frac{g_{m1}}{C_C}$$

$$\frac{10}{C_L} \geq \frac{2.2}{C_C}$$

$$C_C \geq \frac{2.2}{10} C_L$$

$$\boxed{C_C \geq 0.22\ C_L} \tag{5.5.9}$$

So that relation between compensation capacitor and load capacitor is given by in equation (5.5.9). if load capacitor is given, we will easily finds out the value of compensation capacitor